JOB MARKET ANALYSIS REPORT

A data-driven review of job postings, salary patterns, hiring competition, career growth and monthly market movement

100K	■849.95K	81.51
Job postings analysed	Overall average salary*	Average career growth index

Executive Summary

This project analyses a dataset of 100,000 job postings using SQL for data preparation and business queries, Python for data cleaning, and Power BI for an interactive overview dashboard. The analysis compares job categories, cities, company hiring patterns, skill clusters, experience and education requirements, applicant competition, career-growth indicators, deadlines and monthly posting trends.

Key takeaway: The dataset shows broadly similar average salaries across the six job categories, with AI/ML Engineering at the top of the category averages. Gurgaon and Hyderabad lead the city-level average salary comparison, while Chennai has the highest posting count among the cities shown. Tech Mahindra records the highest career-growth index among the companies listed in the query results. Monthly postings fell in February, rebounded in March, and then softened or remained mixed through May.

Dashboard Snapshot

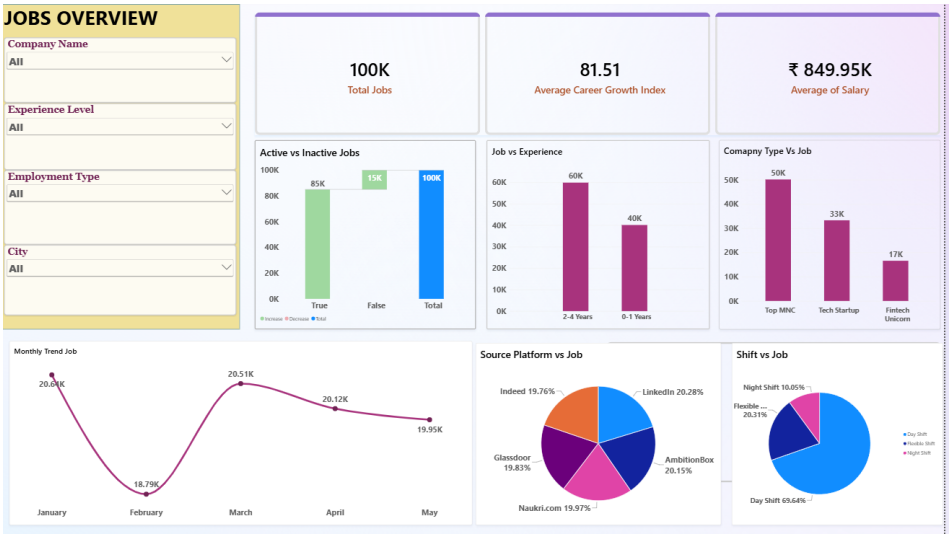


Figure 1. Power BI Jobs Overview dashboard with KPI cards, filters, job status, experience, company type, monthly trend, source platform and shift mix.

1. Project Objective & Approach

The objective was to turn raw job-posting records into clear, decision-oriented insights for job seekers and hiring-market analysis. SQL was used to inspect the table, standardise the posting date, create an average-salary field and answer ten business questions. Python was used to address selected data-quality issues. Power BI brought the results together in a filterable dashboard.

Data preparation performed

• Checked table structure and record count.
• Converted `posted_date` into a DATE field using the supplied day-month-year format.
• Added `average_salary`, calculated as the mean of minimum and maximum salary.
• Filled 33,499 missing values in `probation_period` using the mode.
• Removed the percent sign from `career_growth_index` and converted the column to numeric, coercing invalid entries to missing values.

2. Key Findings

Salary by job category

Job category	Average salary	Postings
AI / ML Engineering	■850,549.94	16,787
Data Science	■850,359.91	16,425
Cloud Architecture	■850,358.51	16,782
Web Development	■849,687.60	16,872
DevOps Core	■849,465.83	16,697
Cybersecurity	■849,267.60	16,437

AI/ML Engineering has the highest average salary in the category comparison (■850,549.94), followed closely by Data Science and Cloud Architecture. The narrow range across category averages suggests that the salary field is relatively similar between categories in this dataset; job-level and city-level differences may be more useful for differentiating opportunities.

City-level salary and opportunity

City	Average salary	Postings
Gurgaon	■852,275.33	12,521
Hyderabad	■852,136.69	12,439
Chennai	■850,362.42	12,636
Bangalore	■850,241.39	12,372
Noida	■849,700.05	12,622

Gurgaon and Hyderabad have the highest average salaries in the supplied city results. Chennai has the largest posting count among these cities (12,636), followed closely by Noida (12,622). Because the posting counts are close, the city decision should consider role fit, cost of living and the actual salary range—not average salary alone.

Hiring patterns and skill clusters

The most frequent company–job-title pairs in the sample results include Cognizant — Cloud Infrastructure Intern (505), Razorpay — AWS Solutions Architect (500), Swiggy — Azure Cloud Engineer (495) and HCLTech — Azure Cloud Engineer (492). These results suggest that cloud-related roles feature prominently in the highest-count combinations returned by the query.

The skills query groups postings by multi-skill clusters rather than counting each individual skill independently. The largest cluster shown is the web-development stack (HTML, CSS, JavaScript, React, Node.js, Express, MongoDB and Tailwind CSS) with 16,872 postings. Other major clusters include AI/ML (16,787), cloud architecture (16,782), DevOps (16,697), cybersecurity (16,437) and data analytics (16,425). These are cluster-level counts and should not be interpreted as independent counts for every skill in a cluster.

Career growth indicators

Among the companies shown, Tech Mahindra has the highest average career-growth index (81.72), followed by HCLTech (81.62) and Paytm (81.56). The dashboard reports an overall average career-growth index of 81.51. This index is treated as a dataset indicator, not a guarantee of an employee's actual promotion or development prospects.

Monthly job-posting movement

The monthly category totals show a dip from January to February 2026, followed by a recovery in March. The strongest March increase in the supplied results is Web Development (+411), followed by DevOps Core (+355). In April, postings softened in several categories, including Web Development (–201) and DevOps Core (–176). May was mixed: DevOps Core rose by 51 and Web Development by 13, while Cloud Architecture declined by 146 and Data Science by 83. The available trend covers January–May 2026 only, so it does not establish a full-year or long-term trend.

3. Dashboard Highlights

The Power BI dashboard provides a concise overview of the dataset through:

- KPI cards for total jobs, average career-growth index and average salary.
- Filters for company name, experience level, employment type and city.
- Active versus inactive jobs, job counts by experience and company type.
- Monthly posting trend, source-platform distribution and shift distribution.

The dashboard is designed to let users filter the market by relevant dimensions rather than relying on one overall number. The displayed shift mix is led by Day Shift (69.64%), followed by Flexible Shift (20.31%) and Night Shift (10.05%). The source-platform distribution is relatively even across LinkedIn (20.28%), AmbitionBox (20.15%), Naukri.com (19.97%), Glassdoor (19.83%) and Indeed (19.76%).

4. Practical Recommendations

For job seekers

- Explore AI/ML, Data Science and Cloud Architecture roles, but compare individual offers and required skills because category-average salary differences are small.
- Include Gurgaon and Hyderabad in salary-focused searches; consider Chennai and Noida for the relatively high posting volumes shown.
- Build a role-aligned portfolio around one skill cluster (for example, Python/SQL/Power BI for analytics, or cloud tools for infrastructure roles) rather than treating the cluster count as proof that every listed skill is equally demanded.
- Check the application deadline and active status together before applying.

For further analysis

- Validate salary units, outliers and whether averages reflect annual compensation consistently.
- Re-run competition ranking after checking applicant-count ties and rank logic.
- Analyse individual skills separately if the source stores several skills in one combined text field.
- Add more months of data before making broader market-growth claims.

5. Data Notes & Limitations

The results are descriptive of the supplied dataset and are not independently verified labour-market statistics. Salary averages depend on the accuracy and units of the source fields. In the applicant-competition output provided, all displayed rows have an applicant count of 479 and a competition rank of 1; this tie means the output does not distinguish those postings by competition and should be validated before drawing conclusions. The deadline output shows 1 February 2026 as expired relative to the current date, including records marked active; such records may need refreshing. Monthly trend results currently cover January through May 2026.

The skills analysis counts combined skill clusters, while the company and city results reflect the query outputs shared for this project. Interpret findings as directional signals within this dataset rather than definitive rankings of the entire job market.

Conclusion

This project demonstrates an end-to-end analytical workflow: data cleaning in Python, structured data preparation and business analysis in SQL, and stakeholder-friendly reporting in Power BI. The analysis surfaces useful comparisons across salary, job categories, cities, hiring combinations, career-growth indicators and monthly posting activity, while also identifying data-quality checks that would strengthen future versions of the analysis.

*Salary values are reported in Indian rupees as presented in the project outputs. Confirm source units and compensation period before external publication.